

Honors Class 08 Activity: What Does “Current” Mean?

Team Members: _____

Twenty minutes, working in the matplotlib warm-up notebook. Predict first, then run. A prediction you got wrong is worth more than one you got right, so write down the wrong one before you fix it.

Part 1: Decode the Format String (5 minutes)

The notebook shows you "-o" and "-ro". Fill in this table **from reasoning alone**, then test each one with `plt.plot(x, y, ...)`.

Format string	Connecting line?	Marker shape?	Color?
"ro"			
"g^"			
"--k"			

Problem 1: Which of your three predictions was wrong, and what rule were you using that turned out to be false?

Part 2: One Cell or Two? (7 minutes)

Put these two lines together in a single cell and run it:

```
plt.plot(x, y1)
plt.plot(x, y2)
```

Now split them: put the first line in one cell, the second line in a new cell below it, and run both.

Problem 2: Before you run the split version — predict. (circle one)

A. Two curves on one plot, same as before B. Two separate plots, one curve each

Now run it. Which happened?

Problem 3: Nothing about the two `plt.plot` lines changed. Only where they live changed. So what does a cell boundary do to a plot?

Part 3: A Table Plot Is a Matplotlib Plot (8 minutes)

The notebook says the datascience table is “calling matplotlib” behind the scenes. Test that claim. In one cell, run `poly.plot("x", ["y1", "y2"])` and then, on the next line of the **same** cell, add a title with `plt.title(...)`.

Problem 4: Did the title land on the table’s plot? What does that tell you about what `poly.plot()` actually is?

Now move the `plt.title(...)` line into a cell of its own, below the `poly.plot` cell, and run both.

Problem 5: Predict first, then run. Describe exactly what appeared — including anything that appeared that you were not expecting.

Problem 6: Write the one-sentence rule you would give a classmate about where plotting code has to live in a notebook, and why `plt.title` needs that rule but `poly.plot("x", "y1")` does not.